POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2023
Programme: BE | Full Marks: 100
Course: Artificial Intelligence | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

a) Justify that "System that think rationally" and "System that act rationally" are the part of artificial intelligence. Explain with examples. 8

b) Define agent function. Differentiate goal-based and utility-based agents with examples. 7

a) What is constraint Satisfaction Problem (CSP)? Solve the cryptarithmetic problem as CSP: WRONG+ WRONG= RIGHT 7

b) Why is Hill Climbing called as Local greedy search? Explain the problems in Hill Climbing with appropriate diagrams. 8

a) Explain the significance of pruning in game playing algorithms. How does alpha-beta pruning work, and how does it improve the efficiency of the search process in minimax algorithms? 7

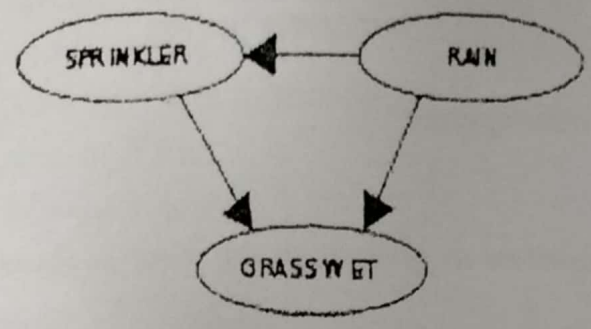
b) Represent the following sentences in first order logic and prolog statements: 8

i. All big cars are expensive.
ii. There are some small cars that are expensive.
iii. A car is expensive only if it is big.
iv. All cars have at least one door.

a) Why do you need to reason in uncertainty? Consider the given Bayesian network with conditional probability tables and find the probability of the events grass is wet, sprinkler is on and it is raining. 7

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	SPRINKLER	
RAIN	T	F
F	0.4	0.6
T	0.01	0.99



	RAIN	
T	F	
0.2	0.8	

		GRASSWET	
SPRINKLER	RAIN	T	F
F	F	0.0	1.0
F	T	0.8	0.2
T	F	0.9	0.1
T	T	0.99	0.01

b) Explain the k-means clustering algorithm with step-by-step demonstration on a small dataset, showing how the clusters are formed. 8

5. a) How does the K- Nearest Neighbor work? Illustrate with a suitable example 8

b) Explain the working principle of support vector machines (SVM) in detail. 7

6. a) What is the role of learning rate in training a neural network? Explain with an example. 8

b) What is fuzzy logic? Explain with its applications. 7

7. Write short notes on: (**Any two**) 2x5

a) Production System

b) Semantic Nets

c) Semantic ambiguity in language processing

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